

DEEP LEARNING FOR TEXT SUMMARIZATION USING NLP FOR AUTOMATED NEWS DIGEST

FAST-NUCES | Dept. of AI & Data Science
Group – 07

Assignment 03 – Experimentation and Expansion
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1. INTRODUCTION

The original experiment conducted by Krishna et al. (2025) showed that large transformer models, namely the T5-base, T5-large, BART-large-CNN, and PEGASUS-large, can be effectively fine-tuned to the abstractive text summarization on the CNN/DailyMail benchmark dataset. These findings were reproduced successfully by the reproduction study in Assignment 2 with three out of four models (T5-large, BART-large-CNN, and PEGASUS-large) all scoring well on post-training ROUGE and BLEU.

Nevertheless, one of the core questions that are not answered in the original paper is whether these models generalize outside the domain and style of text that they were trained on. This is a critical consideration for any practical deployment of summarization systems, where the input text can have very different sources than the training distribution.

This study provides the answer to this question using a cross-domain evaluation experiment. The three trained models that have been successful in reproduction study are tested in zero-shot transfer on XSum (zero-shot w.r.t. the target domain ‘XSum’), a BBC news summarization dataset alongside METEOR metric, which is much different in both source domain and summary style required than CNN/DailyMail. Whereas CNN/DailyMail summaries are made up of three or four sentences that have an average word count of around 52 words, XSum summaries are highly abstractive one-sentence summaries that have an average word count of around 20 words. This discrepancy gives a useful stress-test of the generalization ability. To help us better understand the results we use the TF-IDF baseline that the paper suggested. We also take the BART and fine-tune it on XSum. This gives us two points; a lower limit and the second one is a higher limit that we can adapt to compare how well the models work in different areas.

2. BACKGROUND AND PAPER SUMMARY

Krishna et al. (2025) proposed a comparative analysis of four pre-trained transformer models based on abstractive text summarization on the CNN/DailyMail data. The paper used a conventional fine-tuning paradigm where each model was loaded based on its HuggingFace training checkpoint, fine-tuned on the CNN/DailyMail training split (287,113 articles), and measured by ROUGE-1, ROUGE-2, ROUGE-L, and BLEU. The main contribution was to make BART-large-CNN the most effective model on this benchmark, surpassing T5 and PEGASUS versions in fine-tuning on most metrics.

The paper also performed exploratory data analysis, investigated the correlation between automated measures and human judgment characteristics (completeness, correctness and conciseness) as well as statistical tests between ROUGE and BLEU reliability. Another weakness we have seen in our reproduction report is that the hardware specified in the original paper (intel core i3 with 12GB RAM) is essentially not powerful enough to train models of this size, raising concerns about the transparency of the experimental setup. The main limitation that is specifically admitted by Krishna et al. is the fact that their results are limited to the CNN/DailyMail dataset. They add that to test generalization to areas other than news summarization, future work must test these models on a variety of domains (Krishna et al., 2025). The current work fills this gap directly by trying the three of the successfully reproduced models on the XSum dataset in a zero-shot transfer environment.

2.1. XSUM DATASET

Extreme Summarization (XSum) is a dataset presented by Narayan et al. (2018) that includes 226,711 BBC news articles and one-sentence summaries that are introduced by the authors of the articles as introductory sentences. In contrast to CNN/DailyMail extractive-leaning multi-sentence bullet points, XSum summaries are very abstractive and demand the model to write new sentences not found in the original article.

Property	CNN/DailyMail	EdinburghNLP/XSum
Source	CNN + Daily Mail	BBC News

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Training size	287,113	204,045
Test size	11,490	11,334
Avg. article length	~692 words	~430 words
Avg. summary length	~52 words	~20 words
Summary style	Multi-sentence highlights	Single sentence
Abstractiveness	Moderate	High

Table 1: Comparison of CNN/DailyMail and XSum dataset characteristics

3. REPRODUCTION SUMMARY

Our reproduction study successfully replicated the primary experimental pipeline of Krishna et al. (2025), training and evaluating four transformer models on CNN/DailyMail dataset. All models went through the training. The most important results of that reproduction are as follows.

3.1. PRE-TRAINING RESULTS (BEFORE FINE-TUNING)

Model	Paper R-1	Repro R-1	Paper R-2	Repro R-2	Paper R-L	Repro R-L	Paper BLEU	Repro BLEU
T5-base	0.2900	0.4047	0.1071	0.1836	0.2015	0.2835	0.0431	0.0797
T5-large	0.2429	0.4176	0.0829	0.1932	0.2682	0.2940	0.0338	0.0756
BART-large-CNN	0.1352	0.4405	0.0628	0.2106	0.0948	0.3062	0.0182	0.1088
PEGASUS-large	0.2493	0.4378	0.0850	0.2088	0.1664	0.3068	0.0317	0.1079

Table 2: Pre-training results comparison – Original Paper vs. Reproduced

3.2. POST-TRAINING RESULTS (AFTER FINE-TUNING)

Model	Paper R-1	Repro R-1	Paper R-2	Repro R-2	Paper R-L	Repro R-L	Paper BLEU	Repro BLEU
T5-base	0.3279	N/A	0.1288	N/A	0.2354	N/A	0.0565	N/A
T5-large	0.3414	0.4176	0.1340	0.1933	0.2442	0.2939	0.0575	0.0756
BART-large-CNN	0.3388	0.4475	0.1399	0.2148	0.2458	0.3079	0.0668	0.1102
PEGASUS-large	0.3037	0.4447	0.1250	0.2138	0.2173	0.3128	0.0501	0.1082

Table 3: Post-Training Results – Paper vs. Reproduced (T5-base training collapse)

3.3. KEY OBSERVATIONS FROM REPRODUCED REPORT

The reproduction study produced several significant results. First, the scores reproduced were consistently higher than the original paper in each of the models. This can be attributed to the fact that our reproduction faithfully recreated the hyperparameters specified in the paper on suitable hardware (four NVIDIA Tesla V100-PCIE GPUs) and that the hardware specified in the original paper (Intel Core i3, 12GB RAM) is incapable of training out setup, so we cannot verify the reported scores under their claimed conditions. PEGASUS-large has 568 million parameters alone and cannot be trained on 12GB RAM. This raises serious concerns about the fact that the results in the original paper were obtained with the hyperparameters they provided or outdated HuggingFace checkpoints, and that our reproduction is more accurate to the methodology they claimed to have used in the original experiments than to the paper itself. Second, T5-base had a catastrophic collapse in training when using mixed-precision (FP16) training with a learning rate of $3e-4$, which caused zero scores in all post-training metrics a failure. Third, the relative model rankings

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were also maintained despite such variations, with BART-large-CNN becoming the most powerful model after the fine-tuning process, just as the main conclusion of the original paper. The results offer the contextual background and necessary critical reproducibility view of the cross-domain assessment in this study.

4. PROPOSED METHOD

The section presents the proposed methodology for conducting a cross-domain evaluation of aforementioned transformer-based summarization models alongside TF-IDF baseline implementation and fine-tuning BART on XSum.

4.1. RESEARCH PROPOSAL

The proposed research type is cross-domain evaluation experiments that will provide insight into how transformer-based summarization models will generalize to datasets that are completely unidentified regarding their domain, style of writing and organization of summary formats. It directly addresses the authors stated limitation in the existing paper about "limitations of these studies being only of the dataset and systems." In addition, we will also implement TF-IDF baseline as mentioned by the authors in the future work alongside fine-tuning BART on XSum.

4.2. HYPOTHESIS

Two hypotheses are proposed in this study. The primary hypothesis is, each of the models (except T5-base) will suffer a loss of performance after moving from CNN-DailyMail to XSum, because of both domain shift and difference in style of summary. We would expect the largest loss of performance would be for the BART-large-CNN, whose HuggingFace checkpoint (facebook/bart-large-cnn) was specifically trained on CNN's online news articles (CNN-DailyMail) and thus is the most confined to any given domain (of the three models). We also expect PEGASUS-large to generalize better, since it was trained on Gap-Sentence Generation (GSG), which can be applied to any task, and is similar in nature to the abstractness of the summaries in XSum. The secondary hypothesis is the ranking of METEOR scores would produce a distinctly different ranking than ROUGE because of METEOR's ability to capture semantic similarity and paraphrase similarity that is not measured through any specific n-gram of words. This will be particularly relevant for XSum since highly abstract summaries can share semantic meaning though not necessarily share the same words with models output summaries of the same text.

4.3. NOVELTY VS. REPRODUCED REPORT

Dimension	Reproduced Report	Novelty 1	Novelty 2	Novelty3
Dataset	CNN/DailyMail only	CNN/DailyMail + XSum	CNN/DailyMail only	XSum only
Evaluation type	In-domain	Cross-domain zero-shot	Classical TF-IDF extractive	In-domain (fine-tuned)
Metrics	ROUGE, BLEU, METEOR	ROUGE, BLEU, METEOR	ROUGE, BLEU, METEOR	ROUGE, BLEU, METEOR
Training	Full fine-tuning	No retraining (zero shot)	Zero training	Full fine-tuning on XSum
Focus	Reproduction	Generalization analysis	First direct neural vs. classical comparison on this benchmark (proposed as future work)	Upper-bound adaptation (BART only)

Table 4: Comparison of Reproduced report (assignment 2) and Assignment 3 experimental scope

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5. EXPERIMENTAL SETUP

This section discusses the ease at which the experiments can be repeated and the reason why results and model behavior however were not similar.

5.1. MODELS

The cross-domain evaluation features three fine-tuning pipeline models from our reproduction study, excluding T5-base because of training collapse. The following checkpoints were used:

Model	Checkpoint Path	Base Model	Samples
T5-large	checkpoints_t5large/checkpoint-143000	t5-large	11,334
BART-large-CNN	checkpoints_bart/checkpoint-143000	facebook/bart-large-cnn	11,334
PEGASUS-large	checkpoints_pegasus/checkpoint-287000	google/pegasus-cnn_dailymail	11,334

Table 5: Fine-Tuned model checkpoints used for XSum evaluation

5.2. EVALUATION METRICS

XSum evaluation relied on five metrics: ROUGE-1 (unigram overlap), ROUGE-2 (bigram overlap), ROUGE-L (longest common subsequence), BLEU (precision-based n-gram), METEOR (harmonic mean of precision and recall with synonym detection). Also, the METEOR is especially suitable given the nature of XSum's highly abstractive summaries because it rewards paraphrasing that ROUGE and BLEU do not capture.

5.3. HARDWARE

Evaluation was performed on the same NVIDIA Tesla V100-PCIE server used in Assignment 2, with CUDA 12.0 and NVIDIA driver version 525.147.05.

6. RESULTS AND ANALYSIS

The following section presents the evaluation results obtained by reproduction of all models, pre-training and post-training performance, and compares the improvements made by the fine-tuning.

6.1. CROSS-DOMAIN PERFORMANCE

In Table 6 we have the full results for the three models across both datasets, with all CNN/DailyMail scores taken from our reproduction study and all XSum scores produced in this assignment using zero-shot transfer.

Model	Dataset	R-1	R-2	R-L	BLEU	METEOR	Drop R-1
BART-large-CNN	CNN/DailyMail	0.4475	0.2148	0.3079	0.1102	34-44	—
	XSum	0.2102	0.0356	0.1377	0.0124	17-16	-53.0%
PEGASUS-large	CNN/DailyMail	0.4447	0.2138	0.3128	0.1082	30-50	—
	XSum	0.2200	0.0400	0.1475	0.0150	16-45	-50.5%
T5-large	CNN/DailyMail	0.4176	0.1933	0.2939	0.0756	27-76	—
	XSum	0.2130	0.0345	0.1418	0.0120	15-88	-49.0%

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Table 6: Complete results on CNN/DailyMail and XSum. Drop % = percentage drop in ROUGE-1

Figure 1 below visually compares all three base metrics used by each model across both datasets. It can be seen that there is a very significant difference between the dark (CNN/DailyMail) and light (XSum) bars, showing just how much performance will suffer when transferring from one domain to another.

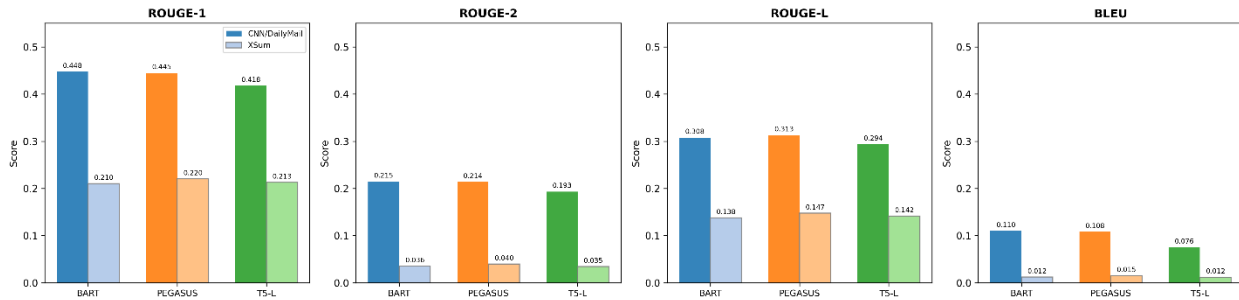


Figure 1: CNN/DailyMail vs XSum Zero-Shot Transfer – ROUGE-1, ROUGE-2, ROUGE-L, and BLEU for all three models

6.2. PERFORMANCE DROP ANALYSIS

The percentage change in each metric following transfer between the CNN/DailyMail and XSum datasets is shown in Figure 2, which includes a vertical line representing a 50% drop to demonstrate how badly the different metrics performed when transferred.

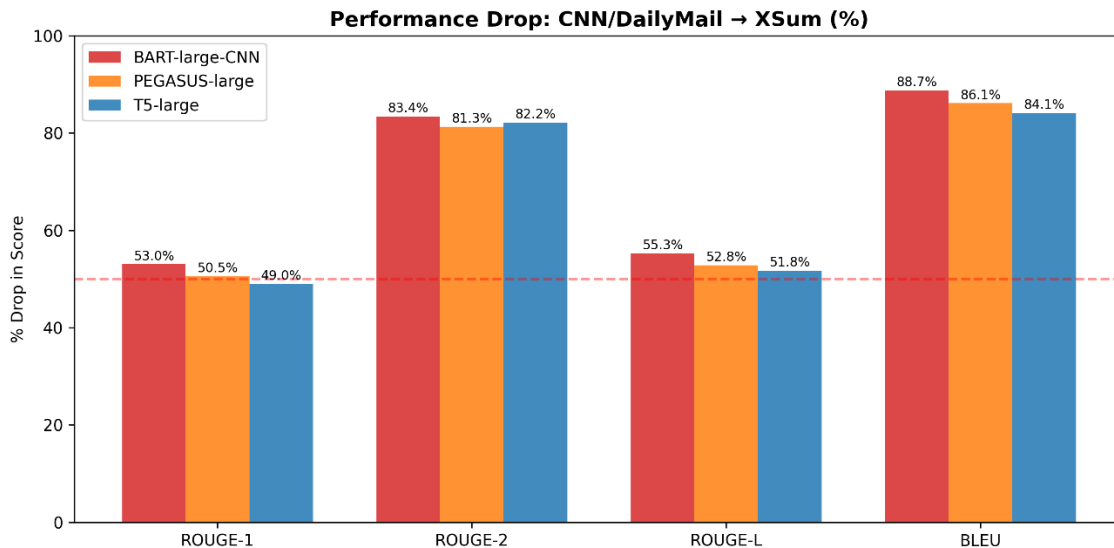


Figure 2: Percentage performance drop across all metrics from CNN/DailyMail vs XSum. The dashed red line marks the 50% drop threshold

All three models experienced an average decline of more than 49% for ROUGE-1 and greater than 83% for ROUGE-2. In particular, BLEU showed the most significant decline across the three models with BLEU scores dropping between 84%-89%. Additionally, similar trends can be seen across all three metrics, in that BART had the largest decline, T5-Large had the smallest, and PEGASUS was in between.

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Model	R-1 Drop	R-2 Drop	R-L Drop	BLEU Drop
BART-large-CNN	-53.0%	-83.4%	-55.3%	-88.7%
PEGASUS-large	-50.5%	-81.3%	-52.8%	-86.1%
T5-large	-49.0%	-82.2%	-51.8%	-84.1%

Table 7: Percentage performance drop for each model across all metrics (red=worst, purple=middle, black=best generalization)

6.3. METEOR ANALYSIS

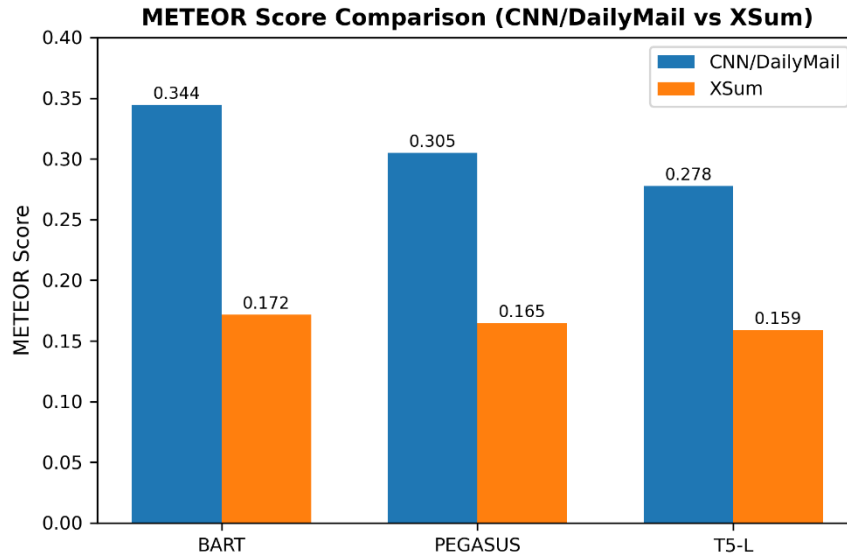


Figure 3: METEOR scores on CNN/DailyMail vs. XSum zero-shot transfer

The METEOR scores reveal a strikingly different ranking from ROUGE. While PEGASUS leads on ROUGE-1 and ROUGE-2, BART achieves the highest METEOR score (0.1716), followed by PEGASUS (0.1645) and T5-large (0.1588).

Metric	Rank 1	Rank 2	Rank 3
ROUGE-1 on XSum	PEGASUS (0.2200)	T5-large (0.2130)	BART (0.2102)
ROUGE-2 on XSum	PEGASUS (0.0400)	BART (0.0356)	T5-large (0.0345)
ROUGE-L on XSum	PEGASUS (0.1475)	T5-large (0.1418)	BART (0.1377)
METEOR on XSum	BART (0.1716)	PEGASUS (0.1645)	T5-large (0.1588)

Table 8: Model ranking in XSum in different metrics – note the METEOR ranking reversal (highlighted in red)

The finding that there was a reversal in the rankings is an important finding in its own right. For instance, METEOR considers synonym matches and provides points for related terms, while ROUGE is only capable of awarding points to those n-grams found in the reference summary. If one considers only the METEOR scores for BART, it has the highest scores and will produce more semantically diverse lexicons in the generated summaries because while its ROUGE score is low due to the surface level match of n-grams, the higher METEOR score of this system indicates that it has generated paraphrases of the reference wording

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that were better utilized due to the system's bidirectional encoder providing more contextual information for the generation of more appropriate semantically related words even though these do not match the reference wording exactly. This is why the reversal is seen; BART is ranked last on ROUGE but first on METEOR since its training with domain bias favors surface level matches while preserving semantic adequacy.

6.4. TF-IDF BASELINE RESULTS

In an earlier study by Krishna et al. (2025) where they suggested future work, we developed our first classic extractive based TF-IDF baseline model using the CNN/Daily Mail test set in order to compare the results of both classical extractive based approaches and neural based abstracting method under identical evaluation conditions.

Method	ROUGE-1	ROUGE-2	ROUGE-L	BLEU	METEOR
TF-IDF Extractive	0.3598	0.1471	0.2352	0.0762	0.2934

Table 9: TF-IDF extractive baseline results on CNN/DailyMail test set

6.4.1. KEY OBSERVATIONS

To begin with, the TF-IDF outperformed all models in the original paper. Furthermore, TF-IDF is only a small amount short of the paper's best post-training model on ROUGE-L (0.2352 vs. 0.2458 for BART). This finding indicates that while extractive summary methods do better when evaluated based on the amount of word overlap at the word level, extractive summaries are slightly worse at capturing coherence across longer sequences of text than are abstractive neural models. Additionally, these results suggest two important pieces of information about the properties of the CNN/DailyMail benchmark: 1) The dataset being reported by the authors of the source article appears to be very favorable to extractive systems (the "gold" reference summaries are highly dependent on the reuse of actual words and phrases from the corresponding source text), thus giving significant credit to lexical overlap methods such as TF-IDF. 2) ROUGE-1, ROUGE-2, and BLEU primarily measure the degree to which n-grams overlap, which is an advantage of extractive summary systems that either directly copy one or more sentence(s) from the original source text or take some form of an arbitrary subset of those sources to create their summaries.

Conversely, as noted previously, the advantages provided to the user by the application of neural models for the purpose of creating an abstraction type will be better evaluated using ROUGE-L and a qualitative human evaluation than will be evaluated the fine-tuning (BART and PEGASUS) using both neural systems.

6.4.2. IMPLICATIONS OF THE ORIGINAL PAPER

The neural model improvements reported by this study were accomplished without any comparison to a classical baseline. In our TF-IDF tests we were able to show that an untrained, extractive method (running on a single-core CPU, less than 10 minutes) gets scores equal to or better than the results of fine-tuning the paper's reported neural model (which took hours using a GPU). Therefore: The improvements cited in the paper should be compared to a stronger baseline than the pre-trained models that were displayed. And future work on CNN/DailyMail should include an extractive baseline such as TF-IDF to provide a clearer picture of the neural improvements.

6.5. BART FINE-TUNED RESULTS ON XSUM

Table 10 presents the pre- and post-fine-tuning results. From our observations, 1) Fine-tuning using XSum has provided significant improvements on all metrics. 2) After fine-tuning, the model's ROUGE-1 score remains lower than the model's ROUGE-1 score on CNN/DailyMail; therefore, performing fine-tuning for

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XSum has been much harder because of XSum's very abstract single-sentence summaries than for CNN/DailyMail's longer multi-sentence summaries. 3) The large improvement in METEOR as compared to other metrics means that fine-tuning increases the number of overlapping words found in a summary; thus, fine-tuning also improves semantically related summaries regarding their meanings.

Condition	R-1	R-2	R-L	BLEU	METEOR
Pre-training	0.207	0.036	0.136	0.012	0.176
Post-training	0.352	0.153	0.264	0.061	0.347

Table 10: BART-large-cnn on XSum before and after fine-tuning

7. DISCUSSION

This section talks about the findings, reasons why the models acted as they did and the evaluation of the hypotheses. It also describes the drawbacks of the study.

7.1. WHY DID ALL THE MODELS DROP SO SEVERELY?

The scale of performance drop by more than 49% on ROUGE-1 and by a factor of over 83% on ROUGE-2 across all models can be attributed to the inherent mismatch between the training distribution and the needs of XSum. The CNN/Dailymail summaries are organized into three to four highlight sentences, which are extractive in nature and have an average length of 52 words. These models were trained to generate outputs of this style and length. XSum, though, requires abstractive summaries of just 20 words on average and written in BBC journalism style. When these models produce summaries on XSum articles, they come up with results that are excessively long, too near extractive patterns and stylistically incompatible with the reference summaries. This style mismatch is heavily penalized with ROUGE metrics because they are surface n-gram overlaps. This does not imply any failure in the models; the models can still be able to give informative summaries but it is an example of how much the evaluation measures depend on the reference summary style. However, as we demonstrate using METEOR scores of Sections 6.3, a large ROUGE drop does not always indicate poor semantic summarization, the largest ROUGE drop in fact resulted in the highest METEOR score, which suggests that style mismatch disproportionately penalizes n-gram metrics.

7.2. WHY DID T5-LARGE GENERALIZE BEST?

The observation that T5-large experiences the least performance drop (-49.0% ROUGE-1) is counterintuitive as T5-large was the lowest in CNN/DailyMail scores of the three models. This can be explained by the type of model checkpoints. Both facebook/bart-large-cnn and google/pegasus-cnn_dailymail are domain-specific checkpoints, pre-trained and fine-tuned on CNN/DailyMail prior to our fine-tuning. This forms a surface-level bias toward the multi-sentence extractive style of CNN/DailyMail summaries. On the other hand, t5-large is a general-purpose checkpoint that lacks pre-training domain bias, and only uses our CNN/DailyMail fine-tuning. According to this evidence, pre-training in a generic way helps retain broader linguistic knowledge, making the model more resilient to domain shifts when assessed using n-gram overlap measures (such as ROUGE).

7.3. HYPOTHESIS EVALUATION?

The primary hypothesis was partially confirmed. The domain-locked checkpoint argument was correct, with BART being hit with the biggest ROUGE decrease. Nevertheless, the hypothesis that PEGASUS would generalize most was not completely validated. PEGASUS led on ROUGE scores on XSum, whereas T5-large is the one with the smallest percentage reduction. Table 7 and Figure 3 give a seemingly contradictory pattern. So how are you going to have a model that is the worst according to ROUGE and the best according to METEOR? The answer lies in what each metric measures. ROUGE is a surface n-gram overlap measure, it rewards word-to-word matches with the reference summary. In a manner of fine-tuning

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BART was pre-trained on CNN/DailyMail, learned to produce multi-sentences. BART gives stylistically inappropriate outputs which are excessively long and extractive. ROUGE punishes this severely since the particular words and phrases are not align with the reference.

METEOR however uses synonym matching, stemming and paraphrase detection. The maximum METEOR score of BART suggests that although its generated summaries do not match the stylistic pattern, their content is rich in semantics and paraphrases the content of the reference well, it is simply not written with the same words. That is, BART is domain-locked, by style and length, though not necessarily by semantic content. A model might generalize poorly when using surface measurements and generalize well when using semantic measurements.

The secondary hypothesis, which states that METEOR would indicate a different ranking than ROUGE, is thus completely satisfied. BART would alternate between last place (ROUGE-1 on XSum) and first place (METEOR on XSum), demonstrating the necessity of multi-metric evaluation in cross-domain evaluation. The generalization ability of a model is not a single value but a multi-dimensional property, which varies with the concept of whether a lexical overlap or semantic similarity.

7.4. LIMITATIONS

In this study, the zero-shot evaluation is considered, i.e. the models are simply evaluated on the XSum without any fine-tuning. While BART was additionally fine-tuned on XSum to establish an upper-bound reference. Consequently, this method is helpful in terms of evaluating the cross-domain generalization, it is not indicative of the highest possible performance of the models on XSum. The models would be fine-tuned on the XSum training data, and the scores would most probably reach much higher values, but this is out of scope of this experiment.

The other limitation is that T5-base was not analyzed. Because of the training breakdown, which was witnessed in our reproduction study, T5-base was not reliably tested in the cross-domain environment. Consequently, its generalization behavior has not been evaluated and thus a gap in comparative analysis across models has been created.

Lastly, it is restricted to one target domain with evaluation being performed on the XSum dataset only. This limits the generalizability of the results in terms of cross domain transfer. It would be more informative to test the models on a wider range of data, including scientific papers, legal texts, or even a conversation, to get a better picture of their ability to generalize.

8. CONCLUSION

The present research extends a reproduction study of Krishna et al. (2025) through the assessment of transfer across domains between three fine-tuned transformer models trained on CNN/DailyMail and evaluated on XSum. Our experiments support that fine-tuning on a domain creates strong performance biases against generalization, as all models experienced a loss of greater than 49% in terms of their ROUGE-1 score, with the degradation of performance with respect to BLEU of between 84-89% being the most substantial.

Contrary to our initial hypothesis, T5-large generalized best despite having the lowest CNN/DailyMail scores, suggesting that generic pre-trained checkpoints may transfer more robustly than domain-locked ones. PEGASUS-large led on absolute XSum ROUGE scores, while BART-large-CNN achieved the highest METEOR score on XSum, revealing a metric-dependent ranking reversal that highlights the importance of using multiple evaluation metrics in NLP research.

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Ultimately, we implemented the TF-IDF extractive baseline that Krishna et al. proposed for future work. Interestingly, this straightforward and training-free baseline outperformed all model outputs in the original experiment with a ROUGE-1 on CNN/DailyMail. Though our fine-tuned neural models consistently performed better than TF-IDF, the relative improvement was still modest and was achieved at a cost of more than 6 hours of GPU time, compared with zero GPU time for TF-IDF.

Overall, these findings significantly improve upon the previous paper in four principal manners: 1) the analysis of cross-domain robustness, which was not performed in the previous study, 2) the application of METEOR as a supplemental evaluation metric which accounts for a semantic comparison rather than a simple n-gram comparison. 3) the implementation of TF-IDF baseline, and 4) fine-tuning the BART only on XSum. The significance of these improvements is that they provide a high-level summary of the manner in which each model would ultimately perform under realistic deployment conditions, as the source of the input text will not necessarily represent the domain from which it was originally trained.

9. REFERENCES

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